

COMPREHENSIVE CHEST X-RAY ANALYSIS

• 8-Zone Analysis • 13 Pilot Findings



Patient ID	-	Study Date	2026-05-19
Name	Sam	Study Time	12:26:46
Age	24	Modality	CHEST X-RAY
Sex	Male	Report ID	cf94cb20

CLINICAL CONTEXT

Indication	Routine screening
Comparison	No prior study available
View	PA (assumed)
Technical Adequacy	Adequate for interpretation

CLINICAL MEASUREMENTS

- Cardiothoracic Ratio: CTR Within Normal Limits (39%)
- Mediastinal Width: Not available
- Hyperinflation: Moderate hyperinflation
- Costophrenic Angle: Blunted
- Hilar Regions: Normal

PRIMARY FINDINGS

Primary Finding: Cardiomegaly
Confidence: 94.0%
Priority: ROUTINE

DIFFERENTIAL DIAGNOSES

Rank	Finding	Confidence
1	Cardiomegaly	49.1%

Note: Each finding is classified independently; probabilities may sum to >100%.

II IMPRESSION

Findings consistent with Cardiomegaly (94% confidence). Localization: Cardiac/Mediastinal. Clinical correlation and radiologist review recommended for definitive interpretation.

II ANATOMICAL 8-ZONE ANALYSIS

Zone	Findings	Status
Right Upper Zone	No significant findings	Normal
Right Mid Zone	No significant findings	Normal
Right Lower Zone	No significant findings	Normal
Left Upper Zone	No significant findings	Normal
Left Mid Zone	No significant findings	Normal
Left Lower Zone	No significant findings	Normal
Cardiac/Mediastinal	Cardiomegaly	Abnormal
Hilar Regions	No significant findings	Normal

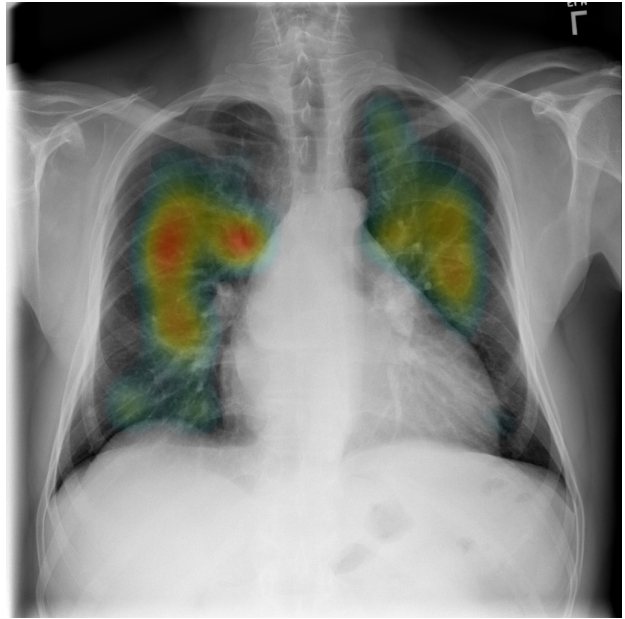
II CLINICAL CLASSIFICATIONS

Classification	Value
ICD-10 Code	I51.7
Cardiothoracic Ratio	39%
Severity	Normal (<50%)
AI Flag vs Measurement	AI flagged cardiomegaly but cardiothoracic ratio is within normal limits — clinical correlation required
Costophrenic Angle	Blunted — possible effusion
Hilar Regions	Normal bilateral hila

AI ANALYSIS VISUALIZATION



Original Image



AI Attention Heatmap

The attention map highlights regions where the AI concentrated its analysis. Warmer colors (red/orange) indicate higher diagnostic relevance.

Attention map reflects the AI's overall focus on this image. Primary finding was reclassified per pilot scope policy — see impression for clinical interpretation.

CLINICAL RECOMMENDATIONS

Priority Level	Primary Action	Follow-up
ROUTINE	Clinical correlation recommended	As clinically indicated
Specialist referral recommended	Cardiology Referral	

IMPORTANT DISCLAIMER

- This AI-generated report is for DECISION SUPPORT ONLY.
- All findings must be verified by a board-certified radiologist.

TECHNICAL APPENDIX

For internal validation / research use — not part of the standard clinical report

Finding Validation Status

Finding	Status	Detection	Evidence
Cardiomegaly	Validated	Ensemble	Internal cohort, 85-92% AUROC
Pleural Effusion	Validated	Ensemble	Internal cohort, 85-92% AUROC
Pneumothorax	Validated	Ensemble	Internal cohort, 85-92% AUROC
Tension Pneumothorax	Supported	Ensemble	Lower-confidence validation subset
Bacterial Pneumonia	Validated	Ensemble	Internal cohort, 85-92% AUROC
Viral Pneumonia	Validated	Ensemble	Internal cohort, 85-92% AUROC
COVID-19 Pneumonia	Validated	Ensemble	Internal cohort, 85-92% AUROC
Pulmonary Edema	Validated	Ensemble	Internal cohort, 85-92% AUROC
Consolidation	Validated	Ensemble	Internal cohort, 85-92% AUROC
Atelectasis	Validated	Ensemble	Internal cohort, 85-92% AUROC
Mass	AI Screening	Vision-language model	Zero-shot, validation pending
Pacemaker Present	Supported	Ensemble	Lower-confidence validation subset
Normal	Validated	Ensemble	Internal cohort, 85-92% AUROC

Per-Model Confidence Breakdown

(Skipped — Per-model probabilities not available (cxr_50path was called without return_details=True).)

Calibration Diagnostics

Metric	Value
Shown confidence (to clinician)	94.0%
Raw ensemble agreement	—
Safety floor (72%) applied	No
Confidence gate triggered	No
Indeterminate flag	False

Attention Map Quality

Metric	Value	Notes
Map type	unknown	luminance/edge composite — not class-discriminative
Entropy	—	Lower = more focused
Coverage (>50% threshold)	—	Fraction of image with above-threshold activation

Peak-to-mean ratio	—	Higher = more pointed attention
Per-pathology localization supported	False	GradCAM B-01 caveat: artifact-fixation on plain CXRs is a known open issue. No per-pathology bounding-box claims are made.

Non-Determinism Diagnostics

Diagnostic	Value
Cache hash method	unknown
Cache hash (sha256-prefix)	—
Repeat-consistency check count	0
Repeat-consistency check pass	—

Study Metadata

Field	Value
Report ID	cf94cb20-a416-41c0-a2b5-96adcb5fb55c
Generated	2026-05-19 12:26:46 UTC
Model name	CXR Ensemble (50-pathway)
Model version	2.0
Model scope	13-pathology pilot scope (10 validated + 2 supported + 1 zero-shot)
Ensemble components	BiomedCLIP, EfficientNet-B4, ResNet50-RadImageNet, MedSAM
Pilot allowlist active	True
Confidence gate triggered	False
Reported timestamp	—
Feedback token	—
PDF code version	v11.59.3